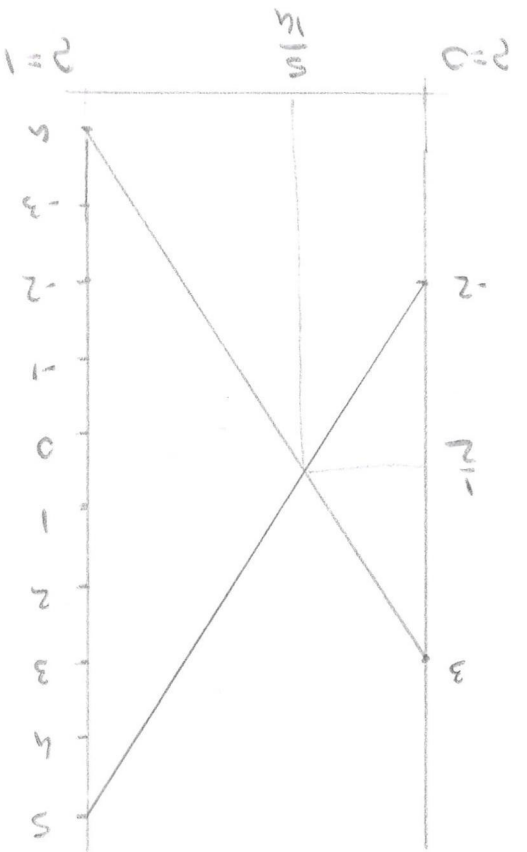


b_1	a_1	a_2
5	5	-2
-4	3	1-2
2		

use graphical method to find value of the game and optimal mixed strategy



expected payoff

$$5p - 2(1-p) = -4p + 3(1-p)$$

$$5p - 2 = -4p + 3$$

$$9p = 5$$

$$p = \frac{5}{9}$$

$$y = 3 - \frac{2}{5} = \frac{14}{5}$$

Player B

$$\begin{bmatrix} -5 & 4 \\ 2 & -3 \end{bmatrix} \begin{matrix} q \\ 1-q \end{matrix}$$

expected payoff: $-5q + 4(1-q)$

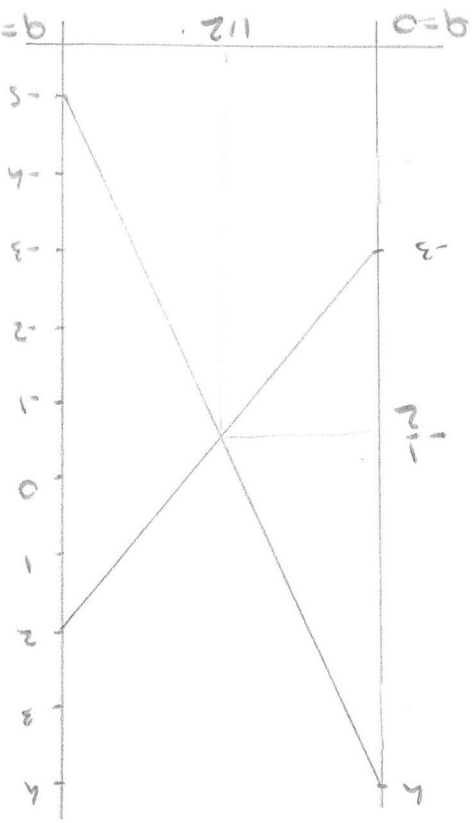
$$-2q + 3(1-q)$$

$$-5q + 4(1-q) = -2q + 3(1-q)$$

$$-9q + 4 = 5q - 3$$

$$q = \frac{7}{12}$$

$$y = 2 \cdot \frac{7}{12} - 3 \cdot \frac{1}{12} = -\frac{1}{2}$$



ASYMMETRIC ZERO-SUM GAME

	b_1	b_2	
a_1	0	4	p_1
a_2	-2	10	p_2
a_3	5	-1	p_3
	$1-q$	q	

EXPECTED PAYOFF B :

$$\begin{aligned} 4q & \quad (a_1) \\ -2(1-q) + 10q & = 12q - 2 \quad (a_2) \\ 5(1-q) - q & = 5 - 6q \quad (a_3) \end{aligned}$$

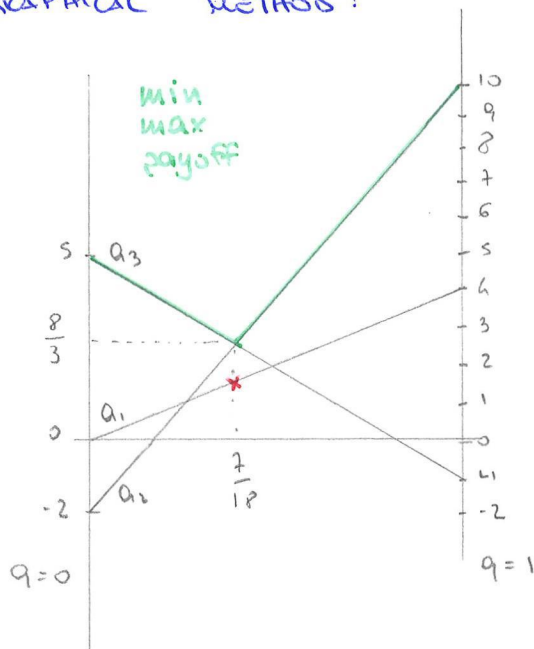
$$4q = 12q - 2$$

$$q = \frac{1}{4} \quad v = 1$$

$$5 - 6q = 12q - 2$$

$$q = \frac{7}{18} \quad v = \frac{8}{3} \quad \leftarrow \text{THIS IS HIGHER!}$$

GRAPHICAL METHOD :



$$\Rightarrow p_1 = 0$$

ALICE WANTS THE MAX-MIN PAYOFF IF SHE WERE TO CHOOSE a_1 IN RESPONSE TO q SHE WOULD GET A LOWER PAYOFF

EXPECTED PAYOFF A :

$$-2p_2 + 5p_3 \quad (b_1)$$

$$10p_2 - p_3 \quad (b_2)$$

$$-2p_2 + 5p_3 = \frac{8}{3}$$

$$-2p_2 + 5p_3 = 10p_2 - p_3$$

$$-12p_2 = -6p_3$$

$$p_2 = \frac{1}{2} p_3$$

$$-p_3 + 5p_3 = \frac{8}{3}$$

$$4p_3 = \frac{8}{3}$$

$$p_3 = \frac{2}{3}$$

$$p_2 = \frac{1}{3}$$

ROCK, PAPER, SCISSORS, LIZARD, SPOCK

SCISSORS CUTS PAPER

PAPER COVERS ROCK

ROCK CRUSHES LIZARD

LIZARD POISONS SPOCK

SPOCK SMASHES SCISSORS

SCISSORS DECAPITATES LIZARD

LIZARD EATS PAPER

PAPER DISPEVES SPOCK

SPOCK VAPORIZES ROCK

(and, as it always has) ROCK CRUSHES SCISSORS

HALF HUMAN, HALF VULCAN

	R	P	S	L	"
R	0	-1	1	1	-1
P	1	0	-1	-1	1
S	-1	1	0	1	-1
L	-1	1	-1	0	1
"	1	-1	1	-1	0

r p s l k probabilities

$$\begin{aligned} \text{EXPECTED PAYOFFS: } & \left. \begin{aligned} -p + s + l - k \\ r - s - l + k \\ -r + p + l - k \\ -r + p - s + k \\ r - p + s - l \end{aligned} \right\} \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} \end{aligned}$$

$$\begin{cases} -p + s + l - k = r - s - l + k & 1 \\ r - s - l + k = -r + p + l - k & 2 \\ -r + p + l - k = -r + p - s + k & 3 \\ -r + p - s + k = r - p + s - l & 4 \\ r + p + s + l + k = 1 & 5 \end{cases} \quad \begin{aligned} -p + 2s + 2l + 2k - r &= 0 \\ 2r - s - 2l + 2k - p &= 0 \\ p + s - 2k &= 0 \\ -2r + 2p - 2s + k + l &= 0 \\ r + p + s + l + k &= 1 \end{aligned}$$

$$\left[\begin{array}{ccccc|c} 1 & 1 & 1 & 1 & 1 & 5 \\ 0 & 0 & 1 & 1 & 2 & 3 \\ -1 & -1 & 2 & 2 & -2 & 1 \\ 2 & -1 & -1 & -2 & 2 & 2 \\ -2 & 2 & -2 & 1 & 1 & 4 \end{array} \right]$$

$$R_3 + R_1$$

$$R_4 - 2R_1$$

$$R_5 + 2R_1$$

$$\left[\begin{array}{ccccc|c} 1 & 1 & 1 & 1 & 1 & 5 \\ 0 & 0 & 3 & 3 & -1 & 1 \\ 0 & 0 & 1 & 1 & -2 & 0 \\ 0 & -3 & -3 & -4 & 0 & -2 \\ 0 & 4 & 0 & 3 & 3 & 2 \end{array} \right]$$

$$R_2 - 3R_3$$

$$\left[\begin{array}{ccccc|c} 1 & 1 & 1 & 1 & 1 & 5 \\ 0 & 0 & 1 & 1 & -2 & 0 \\ 0 & 0 & 0 & 0 & 5 & 1 \\ 0 & 3 & 3 & 4 & 0 & 2 \\ 0 & 4 & 0 & 3 & 3 & 2 \end{array} \right]$$

$$k = \frac{1}{5}$$

$$R_5 - \frac{4}{3}R_4$$

$$\left[\begin{array}{ccccc|c} 1 & 1 & 1 & 1 & 1 & 5 \\ 0 & 0 & 1 & 1 & -2 & 0 \\ 0 & 0 & 0 & 0 & 5 & 1 \\ 0 & 3 & 3 & 4 & 0 & 2 \\ 0 & 0 & -4 & -\frac{7}{3} & 3 & -\frac{2}{3} \end{array} \right]$$

$$R_5 + 4R_2$$

$$\left[\begin{array}{ccccc|c} 1 & 1 & 1 & 1 & 1 & 5 \\ 0 & 0 & 1 & 1 & -2 & 0 \\ 0 & 0 & 0 & 0 & 5 & 1 \\ 0 & 3 & 3 & 4 & 0 & 2 \\ 0 & 0 & 0 & \frac{5}{3} & -5 & -\frac{2}{3} \end{array} \right]$$

$$p = \frac{1}{5}$$

$$\begin{aligned} r &= \frac{1}{5} \\ p &= \frac{1}{5} \\ s &= \frac{1}{5} \\ q &= \frac{1}{5} \\ k &= \frac{1}{5} \end{aligned}$$

STAR WARS VS HARRY POTTER

This game is inspired by the BATTLE of the SEXES, a (very stereotypical) coordination game where a MAN and a WOMAN want to meet in the evening but the MAN wants to go to a PRIZE FIGHT while the WOMAN wants to go to BALLET.

Since I don't like this stereotypical example, I choose some FANDOM !!

TWO FRIENDS WANT TO DO A MOVIE MARATHON. ONE WANTS TO WATCH STAR WARS WHILE THE OTHER WANTS TO WATCH HARRY POTTER. HOWEVER THEY WOULD LIKE TO DO IT TOGETHER. FIND THE EQUILIBRIA.

		F2		
		SW	HP	
F1	SW	3, 1	0, 0	p
	HP	0, 0	1, 3	$1-p$
		q	$1-q$	

EXPECTED PAYOFF F1:

$3q$	(SW)	(depends on F2)
$1-q$	(HP)	

$$\Rightarrow 3q = 1 - q$$

$$q = \frac{1}{4} \quad [\text{PROB F2 PICKS SW}]$$

EXPECTED PAYOFF F2:

p	(SW)	(depends on F1)
$3(1-p)$	(HP)	

$$\Rightarrow p = 3(1-p)$$

$$p = \frac{3}{4} \quad [\text{PROB F1 PICK SW}]$$

GRAPHICAL METHOD:

